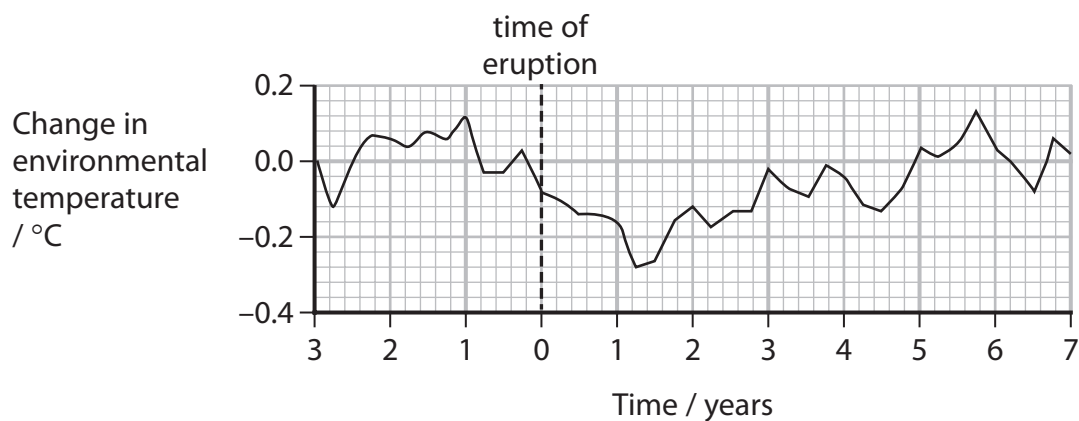


(ii) Explain why succession occurs in stages.

(2)

(b) (i) The graph shows the changes in environmental temperature before and after a volcanic eruption.



Determine the effect of the volcanic eruption on environmental temperature.

Use the information in the graph to support your answer.

(2)

- (ii) When this volcano erupted, sulfur dioxide and particles of ash were released into the atmosphere.

Sulfur dioxide and particles of ash cause less light to pass through the atmosphere.

Explain the results shown in the graph.

(4)

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- (iii) The eruption of one volcano resulted in 5×10^7 metric tons of carbon dioxide entering the atmosphere in one year.

The mass of carbon dioxide released into the atmosphere in one year by anthropogenic activities was 3.5×10^{10} metric tons.

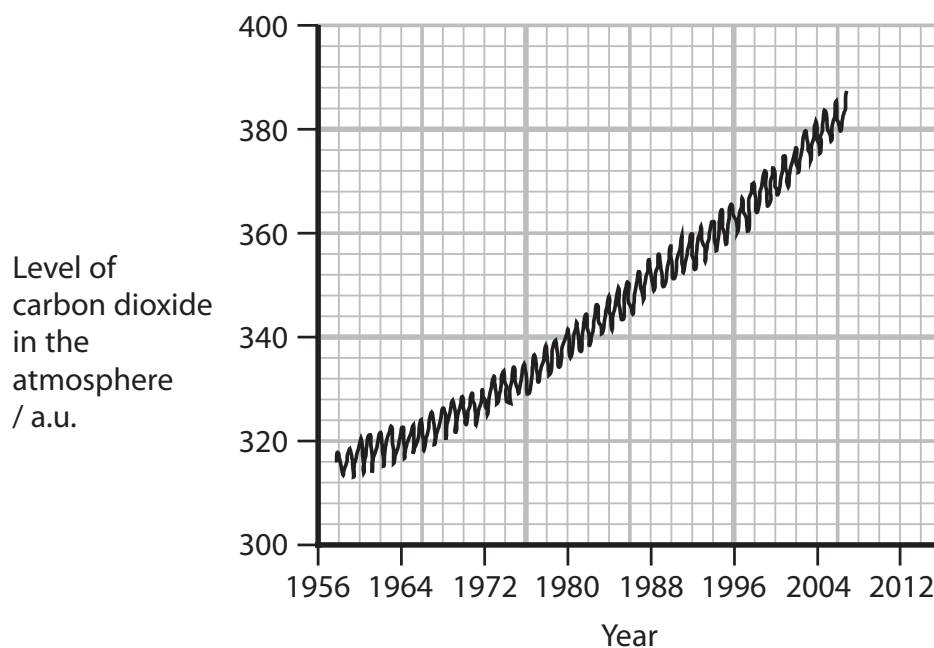
Calculate how many volcanic eruptions would need to occur, in one year, to produce as much carbon dioxide as anthropogenic activities.

(1)

Answer

(iv) The graph shows the levels of carbon dioxide in the atmosphere in one region of the world from 1958 to 2004.

During this period there were volcanic eruptions in 1963, 1980, 1982 and 1991, in this region.



Discuss the possible effects of volcanic eruptions on climate change.

Use all the information in part (b) to support your answer.

(3)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 90 MARKS